

Yannan (Nellie) Wu

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Summary

I am a research engineer at Meta super intelligence labs (MSL), Fundamental AI Research (FAIR) division. My research focuses on system-aware machine learning model architecture design and hardware performance modeling & optimizations. Before Meta, I was an accelerator modeling engineer at Google, working on next-generation TPU performance projection. I obtained my Ph.D. from MIT in computer architecture and systems. I have extensive research experience modeling and designing energy-efficient hardware accelerators for data and computation-intensive applications (such as deep neural networks), in both academic and industrial settings. My works have led to significant contributions to open-source industrial code bases, publications/tutorials at top-tier conferences (e.g., MICRO, ISCA), and a US patent application.

Education

Massachusetts Institute of Technology

Cambridge, MA

Ph.D. in Electrical Engineering and Computer Science (GPA: 5.0/5.0)

June 2023

M.S. in Electrical Engineering and Computer Science (GPA: 5.0/5.0)

Feb. 2020

Advisors: Prof. Joel Emer (emer@csail.mit.edu) & Vivienne Sze (sze@mit.edu)

Cornell University

Ithaca, NY

B.S. in Electrical & Computer Engineering (GPA: 4.02/4.3)

May. 2017

Skills

• C++, Python, C, Verilog, FPGA, PyTorch, MATLAB, HTML, Linux, Git, Docker, Synopsys Design Compiler

Work Experience

- **Meta Research Engineer** Sep, 2023 -
 - Explored hardware-efficient hybrid large language model architecture with space state models.
 - Developed performance modeling flow for datacenter hardware and edge devices.
- **Google Silicon Generalist II** June, 2023 - Sep, 2025
 - Developed analytical modeling infrastructure for various components in TPU chips.
 - Developed large language model training benchmarks in Tensorflow to aid TPU performance modeling.
 - Performed co-design studies to improve TPU compute performance for large language models.
- **NVIDIA Computer Architecture Research Intern** May 2021 - Aug. 2021
 - Explored and analytically modeled various approaches to extend the existing Ampere GPU's sparse tensor core sparsity support to accelerate more structured sparsity patterns and structured sparsity degrees.
 - Developed pruning and fine-tuning procedures using PyTorch to realize various sparsity structures.
 - Filed a patent on a hardware-friendly and novel sparsity structure (US patent application number: 63/236,629).
- **NVIDIA Computer Architecture Research Intern** May 2020 - Aug. 2020
 - Integrated an energy estimation backend (developed at MIT) to NVIDIA's deep neural network accelerator models.
 - Developed a statistical approach to analytically model the energy consumption of various sparse workloads (e.g., pruned deep neural networks).
 - Implemented the proposed modeling flow in C++ and contributed to a large NVIDIA code base.
- **Goldman Sachs Summer Technology Analyst** June 2016 - Aug. 2016
 - Developed filtering functionalities for querying a database of balanced sheets.
 - Developed a web front-end in JavaScript and HTML to allow user-friendly specification of the filter.

Peer Review Service

- Program Committee: 2026 IEEE/ACM International Symposium on High-Performance Computer Architecture
- Journal Reviewer: IEEE Computer Architecture Letters
- Journal Reviewer: IEEE Journal of Solid-State Circuits
- Journal Reviewer: IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems
- Journal Reviewer: IEEE Transactions on Very Large Scale Integration Systems
- Conference Reviewer: ACM Symposium on Parallelism in Algorithms and Architectures
- Journal Reviewer: IEEE Transactions on Circuits and Systems

Teaching Experience

- MIT 6.825 Hardware Architecture for Deep Learning Lead TA Jan. 2020 - May. 2020
- MIT 6.888 Hardware Architecture for Deep Learning TA (part-time) Jan. 2019 - May. 2019
- Cornell ECE 3140 Embedded Systems TA Jan. - May. 2016 & Aug.- Dec 2015
- Cornell ECE 2300 Digital Logic & Comp. Arch. TA Jan. - May. 2015 & Aug. - Dec. 2014
- Cornell MATH 1920 Multivariable Calc. Course Assistant Jan. - May. 2015 & Aug. - Dec. 2014
- Cornell CS 1112 MATLAB Programming Course Consultant Jan. 2014 - May. 2014

Selected Research Experience

- **Software-Hardware Co-design with Novel DNN Sparsity Structures** *MICRO23*
 - Proposed a systematic way to define various structured sparsity patterns used in DNN pruning and proposed a new class of structured sparsity patterns to represent a variety of sparsity degrees.
 - Proposed a novel hardware design methodology to support the proposed structured sparsity patterns with light hardware overhead. Characterized the energy and area of important components with synthesized RTL.
 - Developed pruning/fine-tuning procedures using PyTorch to realize the target sparsity structures in various DNNs.
- **Analytical Modeling of Sparse Tensor Accelerators** *ISPASS21, MICRO22, tutorial at ISCA21*
 - Proposed a taxonomy to systematically describe the previously unstructured and confusing design space of sparsity-related hardware optimizations proposed by existing sparse tensor accelerators.
 - Proposed a decoupled methodology to statistically model sparse tensor accelerators by recognizing the orthogonality between several important design aspects.
 - Developed an *open-source* fast, flexible and accurate modeling framework, **Sparseloop**, to enable design space exploration of sparse tensor accelerators. Contributed >40,000 lines of C++ code to an NVIDIA codebase.
- **Flexible Energy and Area Estimation for Accelerator Designs** *ICCAD19, ISPASS20*
 - Proposed a systematic and flexible methodology to describe accelerator architecture organizations.
 - Based on the methodology, developed **Accelergy**, an *open-source* Python-based infrastructure for architecture-level (pre-RTL) energy and area estimation of accelerator designs.
 - Developed several *open-source* prototype energy and area estimation plug-ins for Accelergy to showcase Accelergy's flexibility to understand user-provided, process-dependent data.
- **Modeling of Fused-Layer DNN Accelerators** *ISPASS23, TCASAI24*
 - Mentored a master student to understand the design space of fused-layer DNN processing.
 - Participated in developing a methodology to systemically describe various fused-layer dataflows.
 - Participated in developing an analytical modeling framework that analyzes the runtime activities of the hardware components in fused-layer accelerators.

Publications and Patent

- **LoopTree: Exploring the Fused-layer Dataflow Accelerator Design Space**
Michael Gilbert, Yannan Nellie Wu, Angshuman Parashar, Vivienne Sze, Joel S. Emer
IEEE Transactions on Circuits and Systems for Artificial Intelligence (TCASAI), Sep. 2024
- **HighLight: Efficient and Flexible DNN Acceleration with Hierarchical Structured Sparsity**
Yannan Nellie Wu, Po-An Tsai, Saurav Muralidharan, Angshuman Parashar, Vivienne Sze, Joel S. Emer
IEEE/ACM International Symposium on Microarchitecture (MICRO), Oct. 2023
- **Accelerating Sparse Tensor Algebra by Overbooking Buffer Capacity**
Fisher Xue, Yannan Nellie Wu, Joel S. Emer, Vivienne Sze
IEEE/ACM International Symposium on Microarchitecture (MICRO), Oct. 2023
- **LoopTree: Enabling Exploration of Fused-layer Dataflow Accelerators**
Michael Gilbert, Yannan Nellie Wu, Angshuman Parashar, Vivienne Sze, Joel S. Emer
IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), April 2023
- **Sparseloop: An Analytical Approach to Sparse Tensor Accelerator Modeling**
Yannan Nellie Wu, Po-An Tsai, Angshuman Parashar, Vivienne Sze, Joel S. Emer
IEEE/ACM International Symposium on Microarchitecture (MICRO), Oct. 2022 (*Distinguished Artifact Award*)
- **Architecture-Level Energy Estimation for Heterogeneous Computing Systems**
Francis Wang, Yannan Nellie Wu, Matthew Woicik, Vivienne Sze, Joel S. Emer
IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), March 2021

- **Sparseloop: An Analytical, Energy-Focused Design Space Exploration Methodology for Sparse Tensor Accelerators**
Yannan Nellie Wu, Po-An Tsai, Angshuman Parashar, Vivienne Sze, Joel S. Emer
IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), March 2021
- **An Architecture-Level Energy and Area Estimator for Processing-In-Memory Accelerator Designs**
Yannan Nellie Wu, Vivienne Sze, Joel S. Emer
IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), April 2020
- **A Systematic Approach for Architecture-Level Energy Estimation of Accelerator Designs**
Yannan Nellie Wu
Master Thesis, Massachusetts Institute of Technology, Feb. 2020
- **Accelergy: An Architecture-Level Energy Estimation Methodology for Accelerator Designs**
Yannan Nellie Wu, Joel S. Emer, Vivienne Sze
IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Nov. 2019
- **Pruning and Accelerating Neural Networks with Hierarchical Structured Sparsity**
Yannan Wu, Po-An Tsai, Saurav Muralidharan, Joel S. Emer
US Patent Application Number: 63/236,629

Conference Tutorials

- **Sparse Tensor Accelerators: Abstraction and Modeling**
Yannan Nellie Wu with Joel S. Emer, Vivienne Sze, Po-An Tsai, and Angshuman Parashar
International Symposium on Computer Architecture (ISCA), June 2021
- **Tools for Evaluating Deep Neural Network Accelerator Designs**
Yannan Nellie Wu with Joel S. Emer, Vivienne Sze, Angshuman Parashar, and Po-An Tsai
IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), Aug. 2020
International Symposium on Computer Architecture (ISCA), June 2020
IEEE/ACM International Symposium on Microarchitecture (MICRO), Oct. 2019

Selected Awards

- MICRO22 Distinguished Artifact Award Oct. 2022
Awarded to ONE paper accepted to MICRO22 based on reproducibility of experimental results
- MIT Jacob's Presidential Fellowship Sept. 2017 - May. 2018
- Cornell ECE Early Career Scholarship June. 2014 - Aug. 2014